

Cloudinary and AWS – Image and Video Storage

1. Cloudinary Documentation

Official documentation

- Main docs:
<https://cloudinary.com/documentation>
- Billing & plans:
https://cloudinary.com/documentation/billing_and_plans
- Credits explained:
https://cloudinary.com/documentation/developer_onboarding_faq_credits

2. How Cloudinary credits work

Free plan

- 25 credits/month

What 1 credit means:

- 1 GB storage OR
- 1 GB bandwidth OR
- 1000 transformations

3. What usage includes

Credits are consumed by:

- Storage (images/videos saved)
- Bandwidth (delivery/viewing)
- Transformations (resize, crop, optimize)

Everything is bundled into one system.

4. Billing behaviour

- Rolling 30-day usage window (not fixed month)
- Free plan resets usage continuously (no carryover of unused credits)

5. Usage Breakdown

Scenario (monthly):

- 1,000 images
- Avg size: 0.5 MB
- 20,000 image views
- 1,000 transformations

Usage Type	Data Amount	Credit Rule	Credits Used	“Cost equivalent”
Storage	0.5 GB	~1 GB = 1 credit	0.5	~\$0.5–\$1 worth of capacity
Bandwidth	10 GB	~1 GB = 1 credit	10	~\$8–\$12 worth of delivery
Transformations	1,000 ops	1000 = 1 credit	1	~\$1–\$2 worth of processing
TOTAL	~10.5 GB usage	—	11.5 credits	~\$10–\$15 equivalent usage value

6. AWS S3 + CloudFront + Lambda Docs

Official documentation

a. S3 (storage)

<https://docs.aws.amazon.com/s3/>

Covers:

- object storage
- buckets
- lifecycle rules
- pricing basics

b. CloudFront (bandwidth/CDN)

<https://docs.aws.amazon.com/cloudfront/>

Covers:

- global CDN delivery
- caching
- bandwidth pricing model

c. Lambda (transformations / compute)

<https://docs.aws.amazon.com/lambda/>

Covers:

- serverless functions
- image processing logic
- request-based billing

7. AWS usage & billing model

a. S3

- Storage → per GB/month
- Requests → per 1000 operations
- Data out (bandwidth) → per GB

b. CloudFront

- Charged for:
 - data transferred (GB)
 - HTTP requests

c. Lambda

- Charged for:
 - execution time (ms)
 - memory used
 - number of requests

8. Usage Breakdown

scenario:

- 1,000 images (0.5 MB each)
- 20,000 views
- 1,000 transformations

Usage Type	Quantity	Pricing Model	Estimated Cost
Storage (S3)	0.5 GB	~\$0.023 per GB/month	~\$0.01
Bandwidth (CloudFront)	10 GB	~\$0.08–\$0.12 per GB	~\$0.8 – \$1.2
Transformations (Lambda)	1,000 ops	~\$0.20 per 1M requests	~\$0.001
Requests (S3/CloudFront)	Low	minimal cost	~\$0.01
Total	—	—	~\$0.81 – \$1.22/month

Note:

- **Bandwidth** = how much data is delivered to users when they view your images/videos.
- **Transformation** = any modification Cloudinary applies to your image/video before showing it.

9. Cloudinary vs AWS S3 Setup Comparison

Usage:

- 1,000 images (0.5 MB each)
- 20,000 views
- 1,000 transformations

Category	Cloudinary (Credits + Price Equivalent)	AWS S3 + CloudFront + Lambda
Storage	0.5 GB $\approx$ 0.5 credits (~\$0.5–\$1 worth usage value)	0.5 GB $\approx$ \$0.01 (S3)
Bandwidth (views)	10 GB $\approx$ 10 credits (~\$8–\$12 usage value)	10 GB $\approx$ \$0.8 – \$1.2 (CloudFront)
Transformations	1,000 $\approx$ 1 credit (~\$1–\$2 usage value)	Lambda $\approx$ \$0.001 – \$0.01
Requests / API	Included in credits	Very small extra cost
TOTAL	11.5 credits (~\$10–\$15 internal usage value)	~\$1 – \$1.5/month real cost

10. Final Summary

Platform	Monthly Cost (same usage)	Real pricing model
Cloudinary	Uses ~11.5 / 25 credits (Free plan OK) OR ~\$10–\$15 equivalent usage value	Credit-based bundled system
AWS Stack	~\$1 – \$1.5/month	Pure pay-as-you-use (separate billing)